



ASSISTIVE HOME CONTROLLER FOR ELDERLY SUPPORT

Pervasive Computing Lab

Final Project Report

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Introduction

The rapid growth of the elderly population has increased the demand for accessible, reliable, and cost-effective monitoring solutions that support independent living while ensuring safety. Advances in embedded systems and wireless communication enable lightweight monitoring platforms that can operate both locally and remotely without complex infrastructure.

In this project, an ESP32-based monitoring system is developed to demonstrate how low-cost microcontrollers can support real-time data acquisition, local decision-making, and remote visualization. By combining device-to-device communication with optional cloud connectivity, the system highlights flexible design choices suitable for home-scale monitoring applications.

The focus of this work is not on medical diagnosis, but on practical system integration, usability for elderly users, and reliable communication across different operational modes. The presented results validate the system as a functional proof of concept with potential for further expansion.

Problem Definition

As populations continue to age worldwide, a growing number of elderly people are living alone or with minimal daily support. While this independence can improve quality of life, it also introduces practical and safety-related challenges. Tasks that seem routine—such as monitoring personal health or reacting quickly to household hazards—may become difficult due to reduced mobility, slower reaction times, or declining sensory awareness. In such situations, even a short delay in responding to an emergency can have serious consequences.

Although many smart home and health-monitoring solutions already exist, they are not always well suited to elderly users. Most commercially available systems assume a certain level of technical familiarity, require complex mobile applications and cloud services to function correctly. In practice, this can make them unreliable in rural areas, frustrating to configure, or simply too expensive for everyday use. From an elderly user's perspective, a system that feels complicated or unpredictable is unlikely to be trusted or used consistently.

Another concern is that many solutions focus on isolated problems rather than everyday realities. For example, a wearable device may track heart rate but fail to communicate effectively with the home environment, while a smart smoke detector may trigger alarms without notifying the user directly or alerting caregivers in a

meaningful way. This separation of health monitoring and home safety reduces the overall effectiveness of assistive technologies, especially in time-critical situations.

The problem addressed in this project is the need for a simple, reliable, and affordable assistive system that supports elderly individuals in their own homes without requiring constant supervision or advanced technical skills. The system should be capable of detecting common but critical events—such as abnormal heart activity, fire or smoke hazards, or unexpected proximity events—and communicating these alerts quickly and clearly.

Proposed Solution

The proposed system introduces an integrated Assistive Smart Home Network composed of two ESP32-based units that communicate directly using ESP-NOW, a lightweight peer-to-peer wireless protocol.

One unit functions as a wearable device (M1) carried by the elderly user and focuses on health monitoring and user interaction. The second unit acts as a stationary home monitoring node (M2) responsible for environmental sensing and hazard detection.

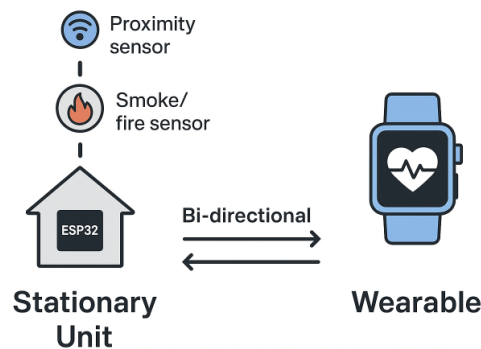


Figure 1: system illustration. [OpenAI Generated Image]

The stationary unit as shown in Figure 1 continuously monitors the home environment using proximity and smoke/fire sensors. These sensors are intended to detect potentially dangerous situations such as fire hazards, gas or smoke presence, or unexpected movement near entry points, particularly during night-time operation. When abnormal conditions are detected, the unit generates an alert and transmits it immediately to the wearable device and onward to caregivers.

The wearable unit focuses on monitoring the user's physiological state through a heart rate sensor. Abnormal heart activity—such as unusually high or low pulse readings—triggers an emergency message that is sent to the stationary unit. From there, the alert can be relayed to caregivers, ensuring that critical health-related events are not confined to a single device.

A key design choice in this solution is the use of bi-directional communication. Both units are capable of sending and receiving alerts, which allows the system to respond flexibly to different emergency scenarios. ESP-NOW was selected specifically because it offers low-latency communication without the overhead of traditional Wi-Fi networking or internet dependency. This choice reduces system complexity and increases reliability, particularly in environments where network connectivity may be inconsistent.

Overall, the proposed solution does not aim to replace full-scale smart home systems. Instead, it provides a focused and practical proof-of-concept that

demonstrates how essential safety monitoring and emergency communication can be achieved using simple hardware, direct wireless communication, and a user-centered design approach tailored to elderly care.

System Architecture

The system architecture is designed around a distributed yet tightly coupled structure that separates environmental monitoring from personal health monitoring. This approach reflects real-world use cases, where risks may arise either from the surrounding home environment or from the user's physical condition. To achieve this, the system is composed of two ESP32-based units that communicate directly using ESP-NOW, enabling fast, local, and reliable data exchange without reliance on external infrastructure.

Stationary Unit (Home Base ESP32)

The stationary unit operates as the home monitoring node and remains fixed within the living environment. It is responsible for detecting environmental hazards and abnormal activity within the home.

Main components and functions:

- Microcontroller: ESP32
- Sensors:
 - Ultrasonic proximity sensor for detecting unexpected movement or intrusion events
 - Smoke / fire sensor for detecting hazardous conditions such as fire or gas presence
- Functionality:
 - Continuously monitoring environmental conditions
 - Evaluates sensor readings against predefined safety thresholds
 - Generates alert messages when abnormal conditions are detected
- Communication:
 - Sends alerts to the wearable unit via ESP-NOW
 - Forwards critical alerts to caregivers through the system's alert mechanism

This unit acts as the first line of defense against environmental risks and ensures that hazards are detected early and communicated without delay.

Wearable Unit (User ESP32 Device)

The wearable unit is designed to be carried or worn by the elderly user, focusing primarily on health monitoring and personal safety.

Main components and functions:

- Microcontroller: ESP32
- Sensor:
 - Heart rate (pulse) sensor for continuous monitoring of the user's physiological state
- Functionality:
 - Continuously measures heart rate
 - Detects abnormal readings that may indicate a medical emergency
 - Generates SOS alerts when thresholds are exceeded
- Communication:
 - Sends emergency alerts to the stationary unit using ESP-NOW
 - Receives environmental alerts from the stationary unit and notifies the user

By keeping health monitored localized on the wearable device, the system ensures immediate detection of personal emergencies without requiring user interaction in critical moments.

Communication and Data Flow

A key architectural decision in this project is the use of bi-directional ESP-NOW communication between both units. Unlike traditional Wi-Fi or Bluetooth-based solutions, ESP-NOW provides low-latency, peer-to-peer communication without network configuration or internet dependency.

The data flow follows two primary paths:

- Health-related Events: Heart rate abnormality → Wearable unit → Stationary unit → Caregiver notification
- Environmental Events: Fire / smoke / proximity detection → Stationary unit → Wearable unit → Caregiver notification

This architecture ensures that alerts propagate quickly and reliably, while maintaining a clear separation of responsibilities between devices. Local alerts are prioritized, followed by escalation to caregivers when necessary, forming a fast and structured response chain.

Required Hardware and Software

Category	Component / Tool	Purpose	Quantity
Microcontrollers	ESP32, ATmega32, Arduino	Main control units	1 each
Communication Modules	HC-05 Bluetooth	Wireless link between master and slaves	2
	Atmega32 flash module	Programming and uploading code to the microcontroller	1
	ESP32 Wi-Fi module	Internet communication (SOS to caregiver)	Built-in
Sensors	MQ-2 Smoke Sensor	Detects fire/smoke	1
	HC-SR04 Ultrasonic Sensor	Intrusion detection	1
	LSM6DSV16BX (IMU)	Fall detection	1
	Pulse Sensor (SE050)	Heartbeat monitoring	1
Actuators / Outputs	Buzzer, LED	Alarm and control outputs	2
Display & Input	Pushbuttons	User acknowledgment	5
Switching Components	Relay	Cut power during fire	1-2
Prototyping Materials	Breadboard, wires, headers	Circuit testing	1 set
Passive Components	Resistors, capacitors, potentiometers	Biasing and filtering	Various
Software Tools	Proteus	Simulation	
	Arduino IDE / Eclipse	Embedded C / MicroPython programming	
	MIT App Inventor / Mobile App	Caregiver SOS interface	
	MS Excel	Gantt chart planning	
	MS Word / PowerPoint	Report and presentation	

Table 1: Required Hardware and Software

Implementation and Results

This project was implemented as a proof-of-concept to validate the feasibility of a simple, locally operating assistive system for elderly care. Rather than focusing on full-scale deployment, the emphasis was placed on demonstrating core functionalities, reliable communication, and practical integration of sensing and alert mechanisms.

The following section presents the outcomes achieved of the project, structured according to the final presentation slides. Each subsection describes a specific milestone or result, supported by figures from the presentation to illustrate the implementation and system behavior.

Before proceeding with hardware assembly, the proposed system concept was first validated through simulation using the Proteus Design Suite. This preliminary step was carried out to verify the feasibility of the overall idea, ensure correct interconnection of components, and reduce potential implementation errors during the physical prototyping phase.

Within the simulation environment, the core controller and peripheral components were modeled to reflect the intended system behavior. Sensor interfaces, communication modules, and basic input–output logic were examined to confirm that signals were correctly processed and that expected responses were produced under different operating conditions. Although the simulation does not fully capture real-world electrical noise or timing constraints, it provided a reliable early indication that the system architecture and control logic were sound.

Conducting this validation step helped clarify design decisions before hardware assembly, minimized the risk of wiring or logic faults, and allowed the development to proceed with greater confidence. Once the concept was verified at the simulation level, the project advanced to physical implementation and incremental testing.

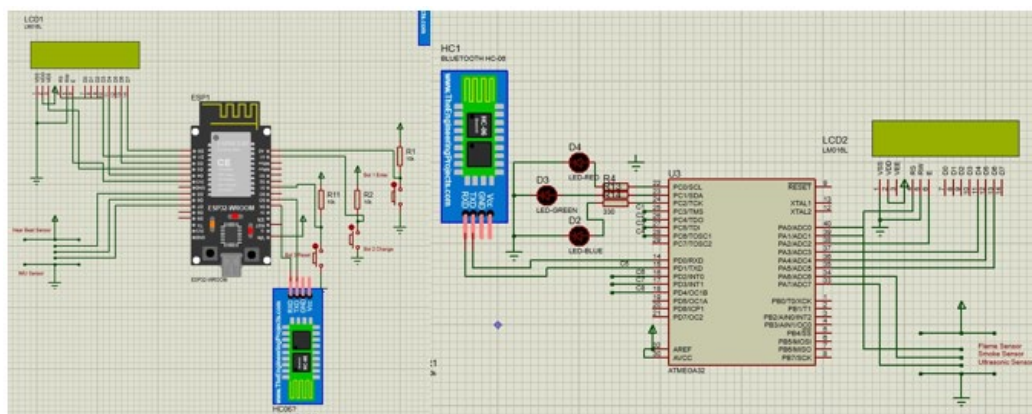


Figure 2: Proteus-based simulation used to validate the system concept prior to hardware implementation.

After validating the system concept through simulation as demonstrated in Figure 2, several practical modifications were introduced during the prototyping phase to improve development reliability and speed. Instead of immediately integrating the LCD

and the three physical push buttons, the Arduino IDE Serial Monitor was used as a temporary interface for both input and output.

This decision simplified early testing and debugging by removing hardware dependencies that could obscure software logic issues. Through the Serial Monitor, three command inputs were implemented to represent three core system functions. These commands allowed the system's navigation logic, state transitions, and alert handling to be tested in a controlled and repeatable manner.

By using the Serial Monitor as a substitute for the physical buttons, user interactions such as selection, confirmation, and reset actions could be simulated through simple character-based inputs. At the same time, system feedback that would normally be displayed on an LCD was printed directly to the serial output. This approach made it easier to trace execution flow, verify correct responses, and refine the control logic before final hardware integration.

Overall, this intermediate step improved development reliability and reduced complexity during early stages, ensuring that the core functionality was stable before transitioning to physical input and display components.

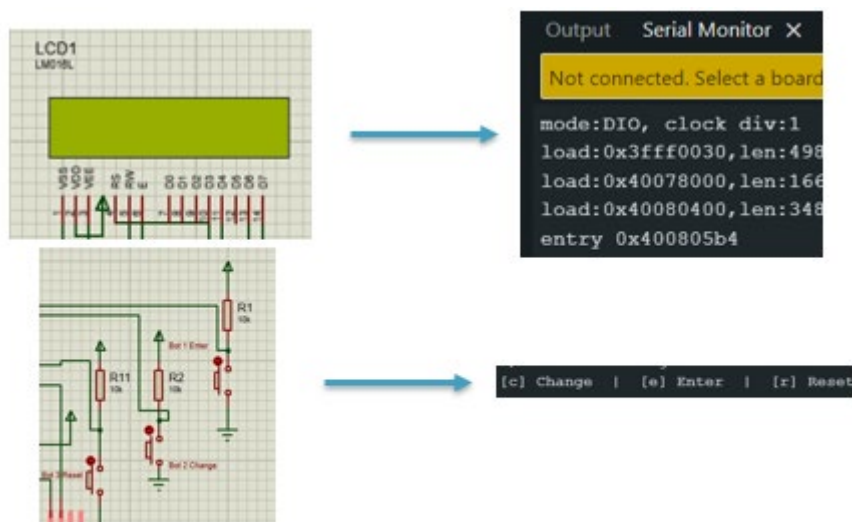


Figure 3: Use of the Arduino IDE Serial Monitor as a temporary input/output interface during prototyping.

To support structured interaction during the prototyping phase, a command-based control mechanism was implemented using the Serial Monitor as in Figure 3. Each user action was mapped to a unique numerical command stored in a global variable (cmd), allowing the system to clearly distinguish between different operational requests.

The following commands were defined and transmitted internally within the system:

- **Send SOS Message:** cmd = 1011

- **Lock the House:** cmd = 1021
- **Option 1:** cmd = 1031
- **Option 2:** cmd = 1041

This encoding approach simplified internal decision-making and ensured that commands could be easily extended or modified in future iterations without restructuring the entire control logic.

Navigation through the system menus was handled via simple character inputs entered through the Serial Monitor:

- **e (ENTER):** Select an option or enter a submenu
- **c (CHANGE):** Cycle through available options
- **r (RESET):** Return to the start screen

The navigation logic followed a hierarchical structure. Pressing ENTER moved the user into a selected submenu, CHANGE allowed scrolling through available options, and RESET provided a direct return to the initial screen at any time. This structure ensured predictable behavior, reduced the risk of user error, and allowed smooth interaction without blocking delays.

By implementing this command and navigation framework, the system achieved a reliable and easily testable interaction model that closely mirrors how a physical interface would operate, while remaining flexible during development.

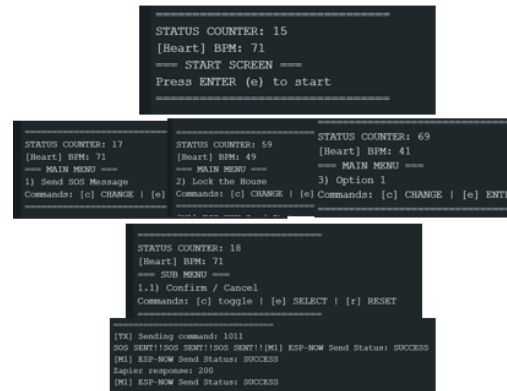


Figure 4: Serial-based hierarchical menu navigation and command execution

After validating the system logic and interaction flow through simulation and serial-based testing as in Figure 4, the project moved to full physical assembly. All required components were integrated into the hardware kits for both the stationary and wearable units, as shown in the assembled prototypes.

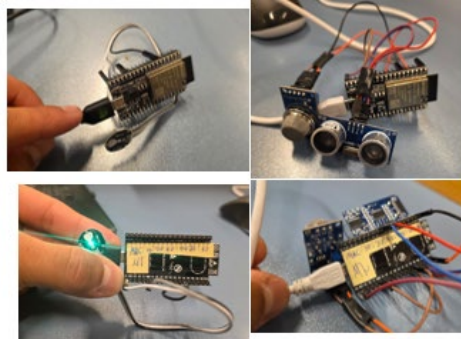


Figure 5: Fully assembled hardware prototypes with labeled ESP32 devices and connected sensors.

Each ESP32 device was clearly labeled with its corresponding MAC address and an identification number as in Figure 5. This step was necessary to ensure correct device pairing and message routing during wireless communication, particularly when using ESP-NOW, where device addressing plays a critical role. Proper labeling also simplified debugging and reduced the risk of configuration errors during testing.

The completed assemblies confirmed correct sensor connectivity, stable power delivery, and reliable operation under real conditions. This stage marked the transition from conceptual validation to a functional hardware implementation, enabling system-level testing of sensing, communication, and alert mechanisms.

Environmental Monitoring (Home Unit – M2)

Environmental monitoring functionality is implemented within the stationary home unit (M2) and is responsible for detecting potential hazards inside the living environment. This unit continuously monitors both proximity and air quality conditions and reacts immediately when abnormal situations are detected.

The data flow for this unit follows a structured pipeline: **Sensors → ESP32 → ESP-NOW → Wi-Fi → Zapier → Google Sheets → Email notification.** This chain ensures that local events detected inside the home are not only handled locally but can also be logged and forwarded to caregivers when necessary.

An ultrasonic sensor is used to measure distance and detect intrusion or unexpected movement within a predefined range. Under normal conditions, the system continuously reports distance measurements without triggering alerts. When the measured distance drops below a defined threshold, the event is classified as a potential intrusion, and an emergency alert is generated.

In parallel, an MQ-based smoke sensor continuously monitors air quality. Smoke levels are evaluated against a predefined safe limit. When the sensor value exceeds this threshold, the system identifies a possible fire or hazardous gas condition and immediately triggers an alert.

In addition to environmental sensing, the home unit is also capable of receiving SOS alerts from the wearable unit (M1). These alerts are generated when abnormal heart-rate conditions are detected on the elderly user's side and are transmitted via ESP-NOW. Upon receiving such a message, the home unit forwards the emergency information through the same notification pipeline.

This implementation demonstrates reliable real-time monitoring, threshold-based decision making, and successful integration of local sensing with remote notification and data logging mechanisms

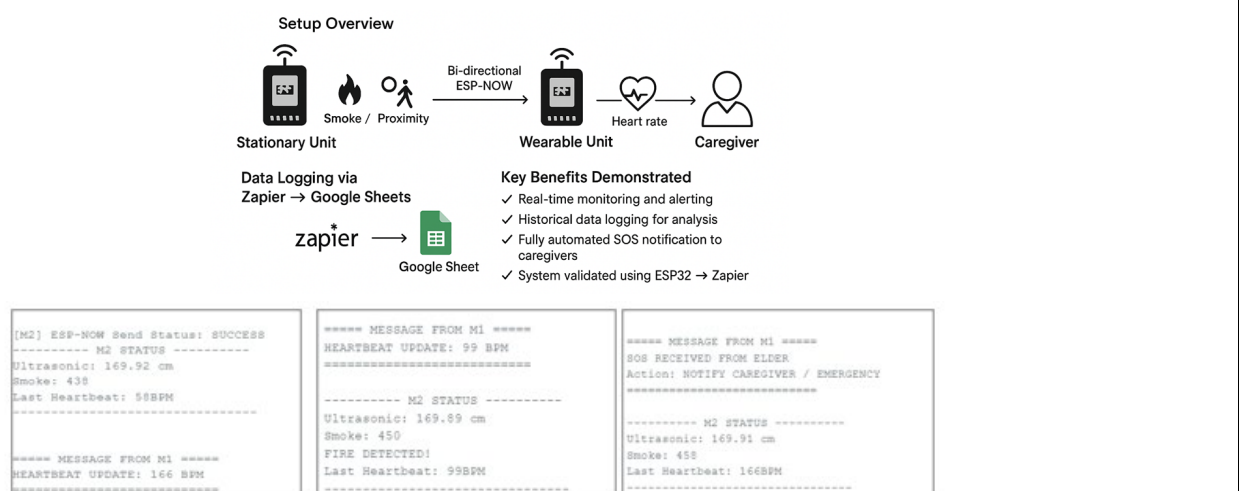


Figure 6: : Environmental monitoring results and alert messages generated by the home unit (M2)

Heart Rate Monitoring (Wearable – M1)

The wearable unit (M1) is responsible for continuous physiological monitoring of the elderly user, with a primary focus on heart rate measurement. A pulse sensor is used to acquire heartbeat signals and calculate the user's beats per minute (BPM) in real time. These measurements are updated continuously and evaluated against predefined thresholds to identify abnormal conditions.

Under normal operation, the wearable unit periodically transmits heart rate data to the stationary home unit (M2). This periodic reporting allows the system to maintain awareness of the user's physiological state without requiring any manual interaction. When abnormal BPM values are detected, the wearable unit generates an SOS alert and immediately transmits it to the home unit.

Communication between the wearable (M1) and the home unit (M2) is fully bi-directional and is implemented using ESP-NOW. This enables low-latency, peer-to-peer data exchange without relying on a Wi-Fi router or external network infrastructure. As a result, critical alerts are delivered reliably and with minimal delay.

Through this communication link, the system successfully transmits multiple types of messages, including:

- Continuous heart rate data
- SOS signals triggered by abnormal heartbeat readings
- Fire alerts originating from the home unit
- Intrusion alerts detected by the proximity sensor

The results demonstrate that the wearable unit can reliably monitor vital signs while maintaining fast and dependable communication with the home unit, ensuring that both health-related and environmental emergencies are handled within a unified alerting framework.



```
===== ESP-NOW MESSAGE RECEIVED =====
FIRE ALERT FROM M2
=====

Zapier response: 200

=====
STATUS COUNTER: 171
[Heart] BPM: 38
=== MAIN MENU ===
1) Send SOS Message
Commands: [c] CHANGE | [e] ENTER | [r] RESET
=====
[M1] ESP-NOW Send Status: SUCCESS
```

```
=====
STATUS COUNTER: 172
[Heart] BPM: 58
=== MAIN MENU ===
1) Send SOS Message
Commands: [c] CHANGE | [e] ENTER | [r] RESET
=====

===== ESP-NOW MESSAGE RECEIVED =====
FIRE ALERT FROM M2
=====
```

Figure 7: Heart rate monitoring and bi-directional alert exchange between the wearable unit (M1) and home unit (M2).

Cloud Integration (Zapier)

To extend the system beyond local monitoring and enable remote logging and notifications, cloud integration was implemented using Zapier. The wearable unit (M1) successfully transmits event data to the cloud by issuing HTTP POST requests to a Zapier Webhooks endpoint whenever a significant event is detected.

Incoming data is received by Zapier in JSON format and processed through a predefined automation workflow. Each event is categorized based on its type and severity, allowing different actions to be triggered depending on the nature of the alert. This design enables flexible handling of multiple event scenarios without modifying the embedded firmware.

- The data sent from the ESP32 to the cloud includes:
- Event type (e.g., fire alert, intrusion, SOS)
- Heart rate value
- Smoke sensor level
- Intrusion distance
- Severity level
- Timestamp

Once received, the data is first logged to a Google Sheets document for record-keeping and later analysis. Conditional filters within Zapier are then applied to determine whether an email notification should be sent to the caregiver. This ensures that only relevant or critical events result in active alerts, while all events remain stored for traceability.

The successful integration of ESP32 devices with Zapier demonstrates reliable cloud communication, structured data handling, and automated alerting. This step validates that the system can scale beyond local operation and support real-time monitoring and notification without requiring a custom backend server.

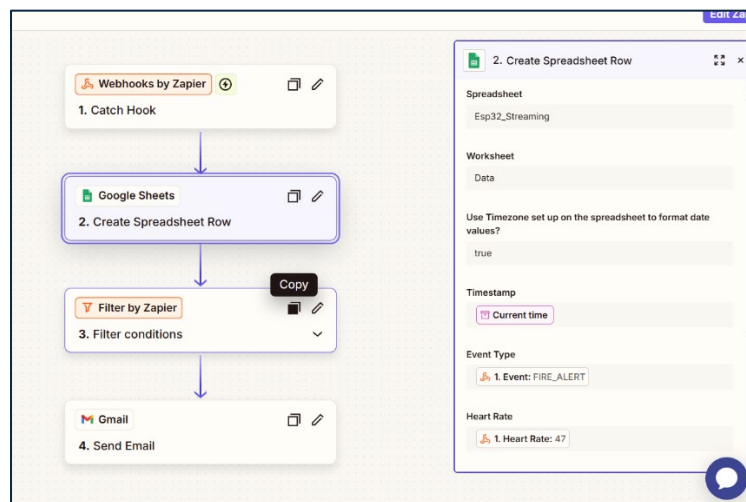


Figure 8: Zapier workflow for receiving ESP32 events, logging data to Google Sheets, and triggering caregiver email notifications

Google Sheets Data Logging

All system data transmitted through the cloud integration pipeline is automatically stored in a Google Sheets document, providing a centralized and persistent data log. This logging mechanism operates continuously and does not require any manual intervention once the system is running.

Normal operating data, such as periodic heart rate measurements, is logged continuously to create a historical record of the user's physiological state. In parallel, emergency events—including fire alerts, intrusion detections, and SOS signals—are recorded immediately when they occur. This ensures that critical incidents are captured without delay and can be reviewed after the event.

Each record is stored with an accurate timestamp generated at the time of cloud reception, enabling precise temporal analysis of both normal and abnormal system behavior. The structured format of the spreadsheet allows easy inspection, filtering, and further processing of the data, supporting both real-time monitoring and post-event evaluation.

By using Google Sheets as a logging backend, the system demonstrates a simple yet effective approach to data persistence and visualization without requiring a dedicated server or database infrastructure.

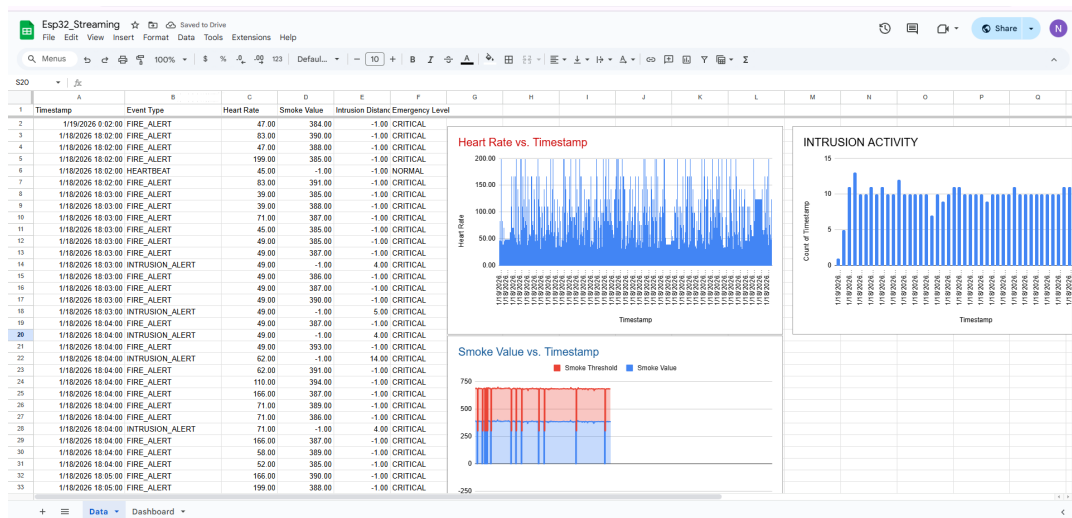


Figure 9: Automatically generated Google Sheets log containing timestamped sensor data and emergency events.

Live Graph Visualization

To enhance situational awareness and enable quick interpretation of system behaviour, live graph visualizations were implemented using the logged data in Google Sheets. These dashboards provide a clear and intuitive overview of both physiological and environmental parameters over time.

Heart rate values are visualized against timestamps, allowing abnormal patterns—such as sudden spikes or prolonged irregular readings—to be identified quickly. Similarly, smoke sensor values are plotted alongside predefined threshold levels, making it easy to recognize hazardous conditions when sensor readings exceed safe limits. Intrusion activity is also visualized through event count graphs, offering insight into movement patterns within the monitored environment.

These visual representations allow caregivers to remotely monitor trends without analysing raw numerical data. At the same time, all historical data remains preserved in the spreadsheet, supporting post-event review and longer-term analysis. The combination of real-time updates and historical context provides a practical balance between immediate awareness and data-driven evaluation.

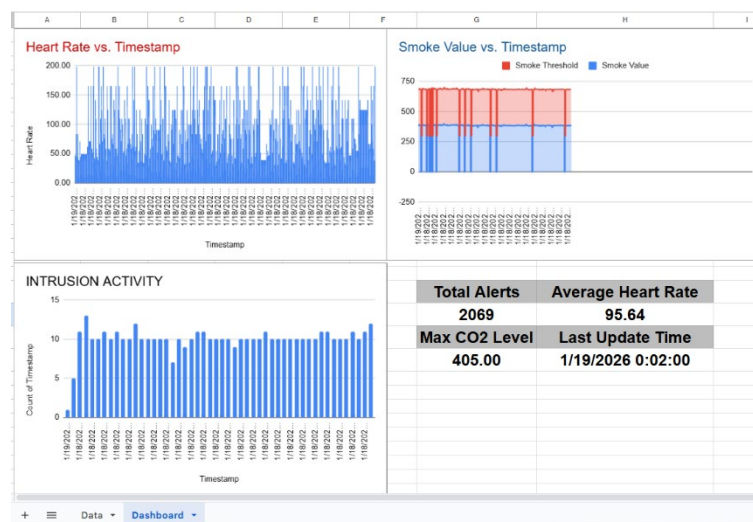


Figure 10: Live dashboard visualization of heart rate, smoke levels, intrusion activity, and system statistics.

Caregiver Email Alert System

To ensure timely human intervention during critical situations, an automated caregiver email alert system was implemented as part of the cloud notification pipeline. Whenever an emergency event is detected, the system automatically generates and sends an email notification to the designated caregiver without requiring any manual action from the elderly user.

Email alerts are triggered for multiple emergency scenarios, including:

- Fire detection when smoke sensor readings exceed the defined safety threshold
- Intrusion detection based on ultrasonic distance measurements
- Abnormal heart rate conditions detected by the wearable unit
- Manual SOS signals initiated by the elderly user

Each alert email follows a clear and structured format to ensure that essential information can be quickly understood under time-critical conditions. The content of the email includes:

- Event type (Fire, Intrusion, or SOS)
- Severity level of the event
- Timestamp of occurrence
- Current heart rate reading
- Smoke level or intrusion distance, depending on the event
- A clear request for immediate action

This structured alerting mechanism allows caregivers to rapidly assess the situation and respond appropriately, even when they are not physically present. The successful delivery of these emails confirms reliable end-to-end integration between local sensing, cloud processing, and real-world notification.

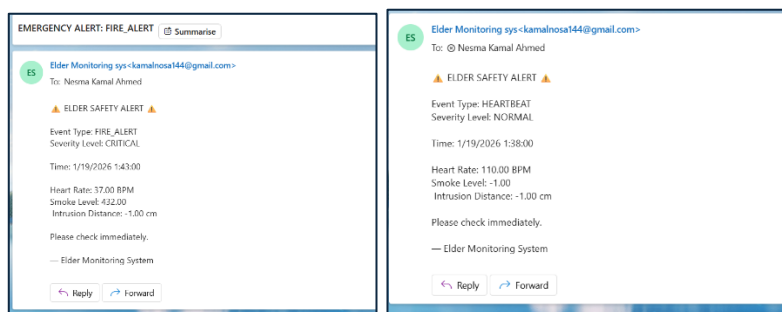


Figure 11: Automatically generated caregiver email alert containing detailed emergency information.

Project Management (Gantt Chart)

Assistive Home Controller for Elderly Support

Course Pervasive Computing Lab

Project Start: Wed, 17/10/25

Display Week: 1

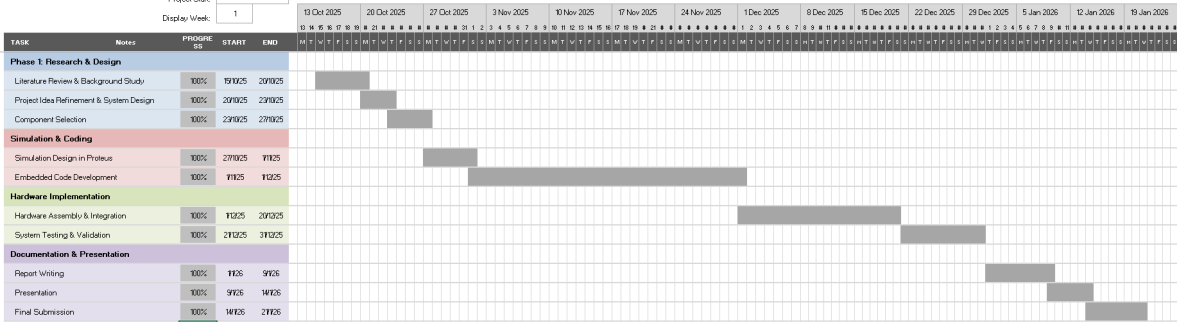


Figure 12: Project Gantt chart [Using Excell]

TASK	Notes	PROGRESS	START	END
Phase 1: Research & Design				
Literature Review & Background Study		100%	15/10/25	20/10/25
Project Idea Refinement & System Design		100%	20/10/25	23/10/25
Component Selection		100%	23/10/25	27/10/25
Simulation & Coding				
Simulation Design in Proteus		100%	27/10/25	11/11/25
Embedded Code Development		100%	11/11/25	11/12/25
Hardware Implementation				
Hardware Assembly & Integration		100%	11/12/25	20/12/25
System Testing & Validation		100%	21/12/25	31/12/25
Documentation & Presentation				
Report Writing		100%	11/1/26	9/1/26
Presentation		100%	9/1/26	14/1/26
Final Submission		100%	14/1/26	21/1/26

Figure 13: Project tasks. [Using Excell]

Discussion

This project demonstrates a functional and low-cost elder monitoring system based on ESP32 microcontrollers, combining local communication (ESP-NOW) and cloud-based services (Wi-Fi, Zapier, Google Sheets). The system successfully achieved end-to-end data flow starting from sensor acquisition, local processing, wireless transmission, cloud logging, and real-time visualization.

The results show that ESP-NOW communication between ESP32 devices is fast, reliable, and energy-efficient, making it suitable for local interactions that do not require internet connectivity. This is particularly valuable for safety-critical actions (e.g., SOS, local alerts), where low latency and independence from Wi-Fi availability are essential.

The Wi-Fi + Zapier + Google Sheets pipeline proved effective for rapid prototyping and visualization. Data was automatically logged and visualized without the need for a custom backend or mobile application. This approach demonstrates that meaningful monitoring dashboards can be built using readily available tools, reducing development complexity and cost.

The menu-based user interface enables elderly users to interact with the system in a simple and intuitive way. Beyond emergency SOS, the inclusion of multiple local actions improves usability and autonomy, allowing the system to support daily monitoring rather than acting only as a panic-response device.

Overall, the experimental results confirm that the system is technically feasible, scalable for small deployments, and suitable for proof-of-concept elder care applications.

Conclusion and Future Work

Conclusion

This project successfully implemented an ESP32-based elder monitoring system capable of local communication, cloud-based data logging, and real-time visualization. The system demonstrates that low-cost embedded hardware, combined with simple communication protocols and cloud tools, can provide effective monitoring solutions without complex infrastructure.

The results validate the system's ability to:

- Collect and transmit sensor data reliably.
- Enable user interaction through a simple menu interface.
- Support both local emergency actions and remote monitoring.
- Visualize data clearly for caregivers using Google Sheets dashboards.

- The project fulfils its objectives as a proof of concept, showing strong potential for real-world elder care applications.

Future Work

Several improvements can be explored to enhance the system:

- Enhanced Menu Functionality: Add adaptive menus, voice prompts, or haptic feedback to further simplify interaction for elderly users.
- Advanced Data Analytics: Implement local anomaly detection (e.g., inactivity patterns, fatigue trends) directly on the ESP32 or a local gateway.
- Mobile or Web Application: Develop a custom dashboard for caregivers instead of relying on third-party tools.
- Improved Security: Introduce encrypted communication, authentication mechanisms, and secure data storage.
- Scalability Testing: Extend the system to support multiple elders and caregivers within the same network.

Reference list

- [1] A. M. Cook, "Ethical issues related to use/non-use of assistive technologies," *Developmental Disabilities Bulletin*, vol. 37, no. 1, pp. 127–152, 2009.
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